LECTURER NOTES

2.2.1.1 Particle density

Particle density measures the density of soil particles only. The measurement of particle density does not include the mass of water or pore space. The formula for calculating particle density is:

Particle density (ρs) = Mass of solids (*M*s) /Volume of solids (*V*s)

Particle density of most mineral soils is about 2.6–2.7 g/cm3. This is because the dominant soil minerals which include quartz, feldspars, micas and clay minerals have an approximate average density of 2.65 g/cm3.

2.2.1.2 Bulk density

Bulk density measures the physical 'looseness' or 'compaction' of a soil. Dry bulk density is the oven dry weight of soil per unit volume. The dry bulk density expresses the ratio of the mass of solids to the total soil volume while total bulk density is an expression of the total mass of a moist soil per unit volume. Note that bulk density is always smaller than particle density.

Dry bulk density (ρb) = *M*s / *V*t

ρb = *M*s / (*V*s + *V*a + *V*w)

Total bulk density (ρt) = *M*t/*V*t = (*M*s + *M*w) / (*V*s + *V*a + *V*w)

Where *M*s = Mass of solids, *M*w= Mass of water, *M*t = Total mass of soil, *V*t = Total volume of soil, *V*s = Volume of solids, *V*a = Volume of air, *V*w = Volume of water

Total bulk density depends more strongly than does the dry bulk density on soil water content. **An ideal soil has a bulk density of about 1.25 g/cm3**. Compacted clay soils have very high bulk densities of between **1.6 and 1.8 g/cm3**.

2.2.1.3 Porosity

Pore space is needed for adequate oxygen supplies to roots, decomposition of organic matter and for storage and infiltration of water. Porosity is an index of the relative pore space in a soil. It is the percentage of a soil that is pore space or voids. Porosity is given by:

Scanned with CamScanner

Porosity (f) = Volume of voids (Vf) / Total volume of soil (Vt)
= (Va + Vw) / (Vs + Va + Vw)
Where: Vs = Volume of solids, Va = Volume of air, Vw = Volume of water
Porosity values generally range from 0.3 to 0.6 (30–60 %). Coarse-textured soils tend to be less porous than fine-textured soils. If both bulk density and particle density are known, the total porosity can be calculated using the formula:
Total porosity = Particle density (ps) – Bulk density (pb) x 100 %
Particle density (ps)

2.2.1.4 Void ratio

The void ratio is also an index of the fractional pore space in relation to the volume of solids. Void ratio is given by:
Void ratio (e) = Volume of voids (Vf) / Volume of solids (Vs)
= (Va + Vw) / (Vt – Vf)
Where Va = Volume of air, Vw = Volume of water, Vt = Total volume of soil, Vs = Volume of solids, Vf =Volume of voids

2.2.1.3 Soil water content

The water content of a soil can be expressed on mass or volume basis.

2.2.1.4.1 Gravimetric moisture content (mass wetness)

Gravimetric moisture content refers to the mass of water relative to the mass of dry soil particles.
Gravimetric moisture content (ɵg) = Mass of water (Mw)/Mass of solids (Ms)
= Mass of wet soil – Mass of oven dry soil
Mass of oven dry soil
Gravimetric moisture content is sometimes expressed as a decimal fraction but more often as a percentage.

2.2.1.4.2 Volumetric moisture content (volume wetness)

Volumetric moisture content (ɵv) = Volume of water (Vw)/ Total volume of soil (Vt)
Thus;
ɵv = Volume of water (Vw) / [Volume of solids (Vs) + Volume of voids (Vf)] 52

Scanned with CamScanner

The volumetric water content refers to the volume of water relative to the total soil volume. At saturation, the volumetric moisture content is equal to the porosity.
Note that volumetric moisture content = gravimetric moisture content x bulk density

2.2.1.5 Degree of saturation

The degree of saturation expresses the water volume present in the soil relative to the pore volume. It is given by;
Degree of saturation (s) = Volume of water (Vw) / Volume of voids (Vf)
= Volume of water (Vw) / [Volume of air (Va) + Volume of water (Vw)]
The degree of saturation ranges from zero in a completely dry soil to 100 % in a saturated soil.

2.2.1.6 Air-filled porosity

Air-filled porosity is a measure of the relative content of air in the soil. It is the volume of air-filled pore space in a soil at given moisture content. This parameter is an important criterion of soil aeration. Air-filled porosity is negatively related to the degree of saturation thus;
Air-filled porosity (fa) = porosity (f) – degree of saturation (s)
Air-filled porosity is also given by;
$fa = Va/Vt = Va/(Vs + Va + Vw)$

(1) Calculate the porosity of a 250 g. Sample that contains 65g of water when 55% of the pores are full of water

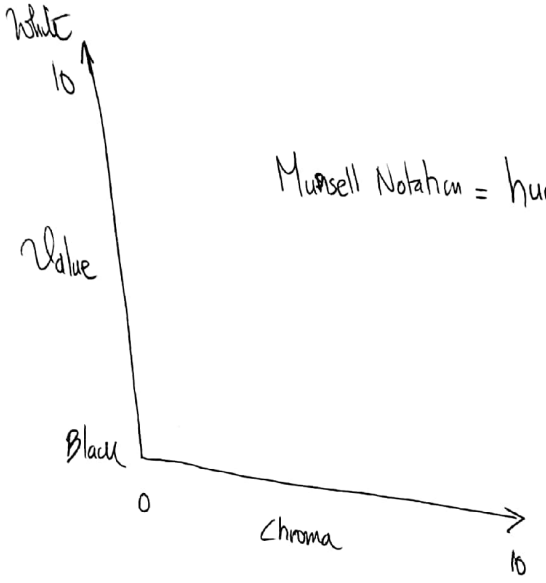
(2) What is the particle density of a Soil Sample that has a bulk density of 1.55 g/cm^3 and porosity of 40%

$$\text{Porosity} = \left(1 - \frac{\rho_b}{\rho_d}\right) \times 100$$

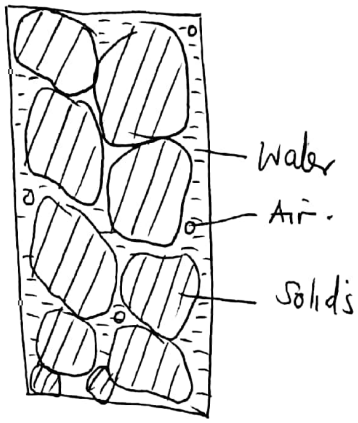
$$40 = \left((1 - 1.55/\rho_d\right) \times 100$$

$$\rho_d = 2.58 \text{ g/cm}^3.$$

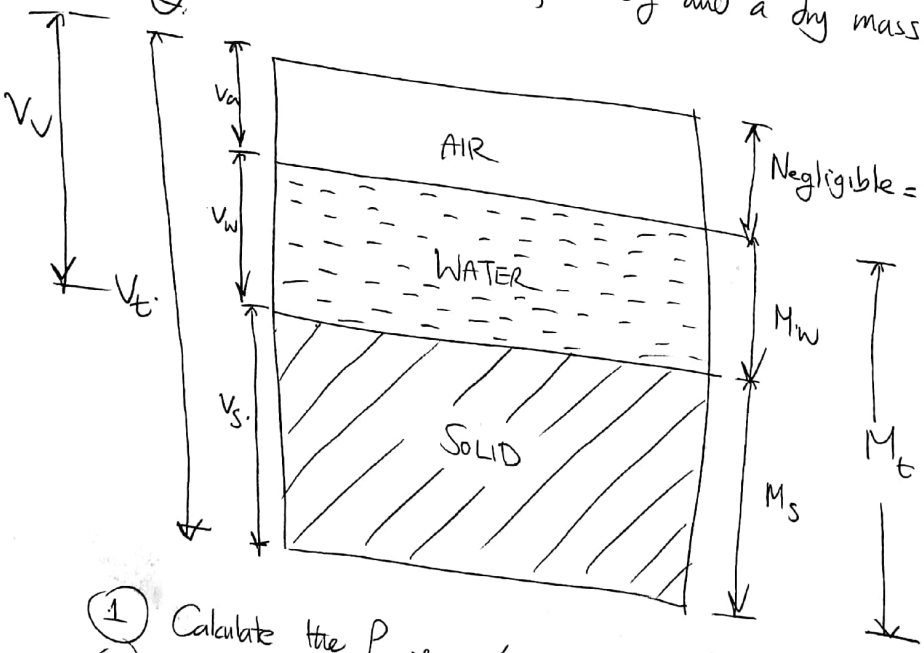
Scanned with CamScanner



Munsell Notation = hue-value-chroma.



(1) Calculate bulk density, porosity, gravimetric water content and volumetric water content for a cylindrical soil sample 10.0 cm long and 7.5 cm in diameter, with a wet mass of 685 g and a dry mass of 572 g.



(1) Calculate the ρ_b of a 400 cm³ soil sample that weighs 575 g (oven dry wt)

(2) Calculate the ρ_b of a 400 cm³ soil sample that weighs 600 g and that is 10% moisture.

(3) Calculate the volume of soil sample that is 12% moisture, weighs 650 g and has a $\rho_b = 1.3$ g/cm³

(4) Calculate the ρ_b of a rectangular soil sample with dimensions 12 cm by 6 cm by 4 cm, that is 15% moisture content and weighs 320 g.

✓ (5) Calculate the porosity of a soil sample that has a $\rho_b = 1.35$ g/cm³ Assume particle density = 2.65 g/cm³

✓ (6) Calculate the porosity (n), of a 250 cm³ clod that contains 140 cm³ water when saturated

Scanned with CamScanner